

CoRoT-2b

R-2118

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_120926 · created 2026-08-11T00:57:59+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2456196.6604 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.206 +/- 0.054
Transit depth (Rp/Rs)^2	4.24 +/- 2.23 [%]
Orbital Inclination (inc)	87.35 +/- 1.96
Airmass coefficient 1 (a1)	1.049 +/- 0.046
Airmass coefficient 2 (a2)	-0.027 +/- 0.042
Scatter in the residuals of the lightcurve fit is	4.4 %
Best Comparison Star	#2 - [175, 214]
Optimal Method	PSF photometry
Transit Duration (day)	0.0916 +/- 0.0061

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.206 ± 0.054 vs 0.1667 (deviation 0.73σ)
Duration fit vs published	0.0916 d vs 0.095 d (deviation 0.56σ)
Mid-time O-C	-6.8 min
Depth SNR	3.8
Residual scatter	4.4 %

Light curve

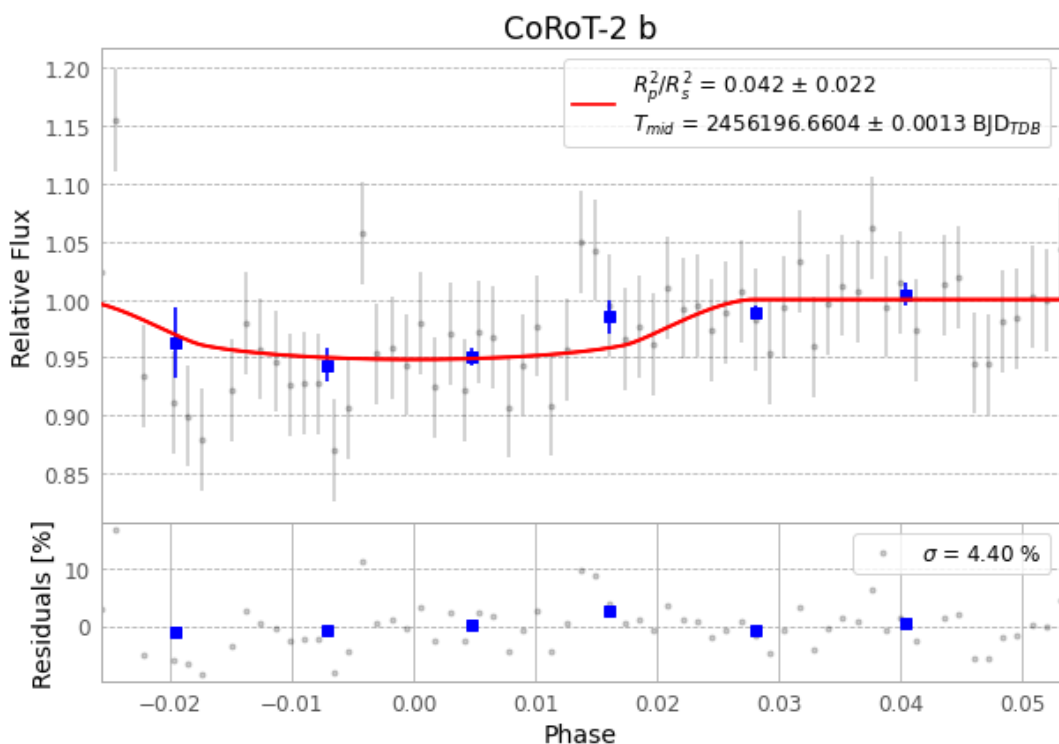


Figure 1 — FinalLightCurve_CoRoT-2 b_2012-09-25.png (SHA-256 cc4f6a8860b5c041...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [310, 274], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0e904e1e366b2074...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

67 FITS frames (44 MB), CoRoT-2120926024213.FITS → CoRoT-2120926060012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-120926.log (137705ad0068...), FinalLightCurve_CoRoT-2 b_2012-09-25.png (cc4f6a8860b5...), FinalLightCurve_CoRoT-2 b_2012-09-25.csv (2b8a66f042c2...)

Compute resources

Wall time 192.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-11 00:58Z.